1.2.6 Wrap-Up: Reflection

1. Rate your level of mathematical confidence on the concepts listed by circling the statement that best describes your understanding.

- **Greatest Common Factor (GCF)**

I don't know what GCF is.	I need a little more practice.	I am okay at this.	I'm pretty good at this.	I could teach this.

- **Least Common Multiple (LCM)**

I don't know what LCM is.	I need a little more practice.	I am okay at this.	I'm pretty good at this.	I could teach this.

Unit 1, Lesson 3: Rewriting Expressions Using the Distributive Property

Benchmark

MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers having a common factor, as a common factor multiplied by the sum of two whole numbers.

Learning Targets

- ☐ I can use the distributive property to generate equivalent numeric expressions.
- ☐ I can rewrite the sum of two composite numbers with a common factor using the distributive property.

1.3.1 Warm-Up: Secret Number

Juanita has a secret number. Read her clues and answer the questions that follow.

Clue 1: My secret number is a factor of 60.

1. What is the smallest value Juanita's number could be?

2. What is the largest value Juanita's number could be?

Clue 2: My number is prime.

3. What values are possible for Juanita's number at this point?

Clue 3: 15 is a multiple of my secret number.

Clue 4: My secret number is a factor of 20.

4. What is Juanita's secret number?

5. Could you have determined her secret number only using the first three clues? Explain.

1.3.2 Guided Instruction: The Distributive Property

The ***distributive property of multiplication over addition*** is a property that applies to all rational numbers. The distributive property "distributes" a factor to all terms within a set of parentheses.

$$a \times (b + c) = (a \times b) + (a \times c)$$

The product 7×23 can be rewritten using the distributive property by decomposing 23 into two parts and multiplying 7 by both parts.

$$7 \times 23 = 7(20 + 3)$$
$$7(20 + 3) = (7 \cdot 20) + (7 \cdot 3)$$

1. What are two other ways that 7×23 could be rewritten using the distributive property?

$$7 \times 23 = \_\_\_(\_\_\_ + \_\_\_)$$

$$7(23) = \_\_\_(\_\_\_ + \_\_\_)$$

2. Which of the expressions is the easiest to evaluate using mental math?

The distributive property can also be used to rewrite the sum of two numbers that share a common factor.

3. Elease rewrote the sum $27 + 18$ as $3(9 + 6)$. Are these two expressions equivalent?

a. $27 + 18$ and $3(9 + 6)$ ○ are / ○ are not equivalent expressions because when evaluated they have
○ the same value.
○ different values.

b. What common factor of 27 and 18 did Elease use to rewrite the expression?

c. Rewrite the expression $27 + 18$ using a different common factor of 27 and 18. Use the area model to support your thinking.

	___ +	___
___	27	18

$27 + 18 = \_\_\_(\_\_\_ + \_\_\_)$

1.3.3 Guided Practice: Rewriting Expressions

1. Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
98×5	$5\,(90 + 8)$	$90 + 8$ 5: 450 | 40	
6×22		___ + ___ ___: |	

2. Complete the table by rewriting each addition expression using the distributive property, modeling with area, and finding the sum.

Addition Expression	Distributive Property	Area Model	Sum
$600 + 140$		$30 + 7$ 20: 600 | 140	
$56 + 32$		___ + ___ ___: |	

1.3.4 Guided Instruction: Rewriting Sums Using Distributive Property

Sums of ***composite whole numbers*** can be rewritten using the distributive property and a common factor of the numbers.

A ***composite number*** is a whole number greater than 1 that has at least one whole-number factor other than one and itself.

1. What are the common factors of 16 and 24?

Factors of 16	
Factors of 24	
Common Factors of 16 and 24	

2. Isaiah rewrote the sum $16 + 24$ using the distributive property. Each step in his thinking is shown. Complete the descriptions of Isaiah's process by examining his thinking and work. Use the words in the bank below to complete each description.

distributive	equivalent	factor	multiplication

Isaiah's Thinking ( ) and Work ( )	Description
"I know 2 is a common factor of both 16 and 24."	Isaiah identified a ___________ both numbers have in common.
$16 + 24 = 2(8) + 2(12)$	He rewrote the expression using ________________ with the common factor.
$2(8) + 2(12) = 2(8 + 12)$	He rewrote the expression using the ______________ property.
"The sum $16 + 24$ can be rewritten as $2(8 + 12)$ using the distributive property."	Isaiah recognized that $16 + 24$ and $2(8 + 12)$ are ________________ numeric expressions.

3. Rewrite the sum $16 + 24$ using the distributive property in two additional ways.

$16 + 24 = \_\_\_(\_\_\_ + \_\_\_)$

$16 + 24 = \_\_\_(\_\_\_ + \_\_\_)$

4. Two students made errors rewriting the sum $54 + 36$ using the distributive property. Each student's expression is shown.

Rewrite $54 + 36$ using the distributive property.	
Naveah's Expression	$4(14 + 9)$
Gavin's Expression	$3(18 + 36)$

Work with your partner to answer the following questions.

a. What error did Naveah make?

b. What error did Gavin make?

c. Generate two different equivalent expressions for the sum $54 + 36$ by rewriting it using the distributive property.

5. Rewrite an equivalent expression for the sum of $40 + 64$ using the distributive property with the **greatest common factor** (GCF).

1.3.5 Collaboration: Missing Values

1. Work with your partner to determine the missing values that make each equation true.

a. $40 + 16 = 4(10 + \_\_\_)$

b. $15 + 80 = 5(\_\_\_ + 16)$

c. $108 + 198 = \_\_\_(6 + 11)$

d. $28 + 63 = 7(\_\_\_ + \_\_\_)$

e. $45 + 18 = \_\_\_(\_\_\_ + 6)$

f. $135 + 27 = 9(\_\_\_ + 3)$

2. Choose one of the equations from question 1 and explain how you determined the missing value(s).

To find the missing value in equation __________, I ____.

3. Have your partner read your explanation and offer suggestions on how to improve it.

1.3.6 Your Turn!

1. Rewrite an equivalent expression for each sum using the distributive property with a common factor.

   a. $51 + 21$

   b. $124 + 20$

   c. $72 + 24$

   d. $39 + 65$

2. Rewrite an equivalent expression for the sum of $68 + 34$ using the distributive property with the **greatest common factor** (GCF).

1.3.7 Wrap-Up: Ticket Out the Door

1. Rewrite an equivalent expression for the sum of $75 + 30$ in two different ways using the distributive property and a common factor.

2. Complete the equation with the missing values that make the equation true.

$$49 + 35 = \_\_\_(7 + \_\_\_)$$